# COMMERCIAL FACILITIES SECTOR DESIGN BASIS FRAMEWORK

*Design Basis, Design Basis Accident, and Military-Action Extension Methodology*
*Function-Primary Facility-Class Architecture — the Mass-Gathering & Convergence-Node Sector*

## DBA-CF: Commercial Facilities Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

**Version 0.5 — DRAFT**

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*For review by engineers, regulators, venue and property operators, and public-safety and policy analysts responsible for the resilience and life-safety of public-assembly, lodging, and retail facilities.*

**SUGGESTED CITATION**

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*This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.*

# Section 1 — Purpose, Scope, and Series Architecture

## 1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Commercial Facilities sector. It is the parent instrument for the DBA-CF series and applies the DB → DBA → DBA-MA three-layer analytical progression to the public-assembly, lodging, and retail functions the sector performs — and to two roles those facilities carry that the conventional soft-target framing omits: their function as disaster-support and mass-care sites, and their frequent position as convergence nodes where several other critical-infrastructure sectors physically stack.

## 1.2 What This Document Does Not Do

This document does not supersede building and fire codes or public-safety authority. It does not restate the Bright Line or Command Broker, inherited by reference from FD-BL. For its consumer character it does not re-scope the physical infrastructure its facilities run on: the energy, communications, transportation, and water they depend on are governed by their own series and inherited here.

## 1.3 Function-Primary Organizing Principle

Placement follows the consequence chain. The sector is organized by function — public assembly and mass gathering, lodging and real estate, and retail commerce — and each function is placed by the life-safety and economic consequence its failure produces, sub-indexed by occupancy and by convergence exposure.

## 1.4 Series Placement — DBA-CF as a Standalone Sector Series

DBA-CF is a standalone sector series and the sixteenth NIPP sector addressed by the corpus. It inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination. The Department of Homeland Security is the Sector Risk Management Agency.

## 1.5 Beyond the Soft Target — Two Roles the Framing Omits

The sector taxonomy characterizes Commercial Facilities as high-visibility, publicly accessible soft targets with a mass-gathering vulnerability, and that is correct as far as it goes: the mass-gathering life-safety consequence — tens of thousands of people in a single venue — is the sector's most visible design basis concern. But two roles that the soft-target framing omits are, for this framework, the sector's most important contributions.

The first is the disaster-support role. Large public-assembly venues are the nation's mass-care infrastructure: when a hurricane, flood, or wildfire forces evacuation, the stadium, arena, and convention center become the emergency shelter and reception center under Emergency Support

Function 6. A commercial facility therefore carries a response-asset consequence that appears precisely when another sector fails — its value as shelter capacity is highest at the moment a disaster is at its worst — and its loss is not merely the loss of a venue but the loss of the shelter the population needs. This couples the sector directly to the emergency-services sector and to the population-protection mission of the whole framework.

The second is the convergence-node role. Large urban commercial facilities are frequently located where two or more other critical-infrastructure sectors are deeply involved within a small radius. Madison Square Garden is the defining example: the arena sits directly above Pennsylvania Station, so within a few blocks the site stacks intercity and commuter rail, subway, surface transportation, the electric supply that serves them all, and the communications that coordinate them. For such a facility the design basis cannot be a single-facility analysis. An event at the node cascades across the co-located sectors, the facility's own resilience depends on all of them at once, and a mass-gathering event places tens of thousands of people atop that convergence. This is the first consequence structure in the corpus organized by geography rather than by a single consequence chain, and it is the natural bridge from the sector frameworks to the cross-sector interdependency analysis.

# Section 2 — The Commercial Facilities Sector and Its Regulatory Context

## 2.1 A Privately-Owned, Code-Governed, Open-Access Sector

Commercial Facilities is almost entirely privately owned, open to the public by design, and governed primarily by local building and fire codes rather than by a federal operating regulator. Accessibility is governed by the Americans with Disabilities Act, worker safety by occupational-safety standards, and gaming, where applicable, by state regulators. The Department of Homeland Security is the Sector Risk Management Agency, and its work on securing soft targets and crowded places is the nearest federal security framework. The sector's defining characteristic is open public access, which is both its purpose and its vulnerability.

## 2.2 Standard-and-Function Inheritance

The applicable standard varies by function: public-assembly venues inherit the occupancy, egress, and life-safety provisions of the building and fire codes and the DHS crowded-places guidance; lodging and real estate inherit the building codes and accessibility requirements; and retail inherits the same, with state gaming regulation where relevant. The disaster-support role additionally invokes the Emergency Support Function 6 mass-care framework, and the convergence-node role invokes the standards of every co-located sector.

## 2.3 The Regulatory Gap

The sector is governed by life-safety codes calibrated to accidental events — fire, structural failure, crowd crush — but carries no design basis for the deliberate mass-casualty attack, for the

loss of its disaster-support capacity when it is most needed, or for the convergence-node consequence in which an event propagates across co-located sectors. Code compliance is measured in occupancy and egress; a design basis measured against deliberate action and cross-sector convergence does not exist.

# Section 3 — Function-Primary Taxonomy

## 3.1 The Function Classes and the Occupancy / Convergence Sub-Index

The sector is decomposed into three function classes — public assembly and mass gathering, lodging and real estate, and retail and commerce — and one coordination class, the venue-security-and-coordination layer, with the disaster-support and convergence-node dimensions carried as overlays. Each is sub-indexed by occupancy and by convergence exposure, so that the design basis load a given facility carries reflects both how many people it holds and how many other sectors it stacks upon.

## 3.2 Public Assembly & Mass-Gathering Venues (PA)

The public-assembly function covers sports stadiums and arenas, performing-arts venues and theaters, convention centers, theme parks, and large outdoor events. Its governing consequence is a mass-gathering life-safety consequence, and it is the class on which both overlays concentrate: it is the disaster-support and mass-care facility of Section 1.5, and it is most often the convergence node. It is therefore the highest-consequence and most analytically distinctive class of the sector.

## 3.3 Lodging & Real Estate (LR)

The lodging-and-real-estate function covers hotels and motels, office buildings, multi-unit residential buildings, and mixed-use developments. Its governing consequence is an occupancy life-safety consequence at a scale below the mass-gathering venue but sustained around the clock, and its high-rise and mixed-use forms carry their own egress and structural design basis.

## 3.4 Retail & Commerce (RT)

The retail-and-commerce function covers shopping centers and malls, retail stores, gaming facilities, and food service. Its governing consequence combines a public life-safety consequence with an economic-hub consequence, and its just-in-time supply and point-of-sale dependencies tie it to the transportation, communications, and financial sectors.

## 3.5 CorpCo — Venue Security & Emergency Coordination (coordination class)

The coordination function is carried by a coordination class spanning private venue and event security and its interface with public safety — the crowd-management, screening, and incident-coordination capability on which a mass-gathering event depends. In the baseline these are

private venues running their own enterprise-internal security, which is why the coordination class is designated CorpCo; when the disaster-support / ESF-6 mass-care overlay activates, coordination of the venue is handed off to Emergency Services (EM). It is the locus at which the AI-Governance Design Basis is implemented for the sector — the Bright Line placement and Command-Broker obligation for machine-learning and generative-AI systems in video surveillance and crowd analytics, access control and screening, and building-management automation.

*† The PA / LR / RT and CorpCo designators were ratified 2026-08-09. The disaster-support (mass-care) and convergence-node dimensions are carried as overlays because they cut across the function classes and connect the sector to Emergency Services and to the cross-sector interdependency analysis respectively.*

### 3.6 Consequence Measures

The sector requires three consequence treatments. The mass-gathering life-safety consequence is measured by occupancy and egress capability under a deliberate-event condition. The disaster-support consequence is measured by the mass-care shelter capacity lost relative to the population that depends on it, calibrated against the disaster that would invoke it. And the convergence-node consequence is not a single-facility measure at all: it is the aggregate cross-sector consequence of an event at the node, assessed against every co-located sector, and it is the sector's formal entry point to the interdependency analysis. *Public Assembly and Mass-Gathering Venues* (DBA-CF-PA-FC1) states the three for the public-assembly function as a life-safety chain, a refuge chain and a node chain, adds crowd density, perimeter reach and stop time to the life-safety measure, states shelter capacity and sustainment as the disaster-support measure, and states the convergence count as the entry point it hands to the interdependency analysis (*Public Assembly and Mass-Gathering Venues* §3.2, §3.3, §10.2, §10.3); this framework takes the statement up at its next conformance pass.

## Section 4 — The Commercial Facilities Sector Design Basis

*[Skeleton — to be developed.] The design envelope will be organized by event category — mass-casualty attack, loss of disaster-support capacity, and convergence-node cascade — with a consequence-severity inclusion criterion consistent with the series.*

### 4.1 Design Basis Principles

The criterion for inclusion is life-safety, disaster-support, or convergence consequence severity — potential for mass-casualty harm at a public gathering, for the loss of mass-care capacity when a disaster requires it, or for a cross-sector cascade at a convergence node — not event likelihood. Candidate bounding events, to be developed: a deliberate mass-casualty attack on a full venue; the loss or denial of a venue in its disaster-support role during an active event; a structural or fire event at a high-occupancy facility; and an event at a convergence node whose consequence propagates across the co-located transportation, energy, and communications infrastructure.

# Section 5 — Dual Design Basis: Bright Line and Command Broker

*[Skeleton — to be developed on the DBA-W §5.5 model.]*

Commercial facilities are heavy adopters of AI in physical security — video analytics, crowd-density and anomaly detection, and automated access control and screening — and in building management. The Bright Line must be placed for any action of a component within scope of FD-LD §4.3 whose incorrect performance could affect crowd life-safety at or above the severity threshold — an automated egress, lockdown, or crowd-flow decision, a screening disposition, or a life-safety-system actuation — with the architecturally enforced authorization of a qualified security officer or incident commander as Command Broker. Determinism decides scope under FD-LD §4.3; it does not decide placement.

# Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

## 6.1 Inward View — and the Convergence Tangle

Commercial Facilities depends on energy for electricity and HVAC, on communications for point-of-sale and security systems, on transportation for customer access and supply delivery, and on water for sanitation and food service. For the convergence-node facilities of Section 1.5 these dependencies are not a clean set of separable chains: the transportation the venue depends on is physically the same infrastructure the venue sits upon, the electric supply is shared across the node, and a disruption to any co-located sector is simultaneously a disruption to the facility. The convergence node's inward view is a tangle, and its design basis must treat the co-located sectors as a coupled system rather than as independent dependencies.

## 6.2 Outward View — Life-Safety, Economy, and the Response Reserve

The outward consequence of the sector runs in three directions: the immediate life-safety consequence of a mass-gathering event, the economic consequence of the sector's role as an economic-activity hub, and — distinctively — the response-reserve consequence, in which the loss of a venue's disaster-support capacity degrades the emergency-services and population-protection mission of the whole framework at the moment it is most needed.

## 6.3 Named Boundaries

Three boundaries are named. The disaster-support role is a shared interface with Emergency Services: the facility and its life-safety systems are Commercial Facilities, while the mass-care sheltering mission it hosts is governed by the emergency-services and Emergency Support Function 6 framework. The convergence-node role is a shared interface with every co-located sector — most often transportation, energy, and communications — and it is the point at which the DBA-CF design basis hands off to the cross-sector interdependency analysis that the corpus maintains separately.

The third is the planned gathering, a shared interface with Emergency Services stated from both sides on the author's decision of 7 October 2026, with the line drawn by who acts. The venue's and the promoter's side belongs to this series, in the public-assembly function and the CorpCo venue coordination class: medical capacity and supplies at the venue sized for the mass-casualty surge the event could produce rather than for an ordinary event day, and the duty to bring every responding discipline — law enforcement, fire and rescue, and emergency medical services — into the event's plan before it opens. The response's side belongs to the Emergency Services series: a plan joining every discipline, a staffed dispatch position for the event, and check-in for the responders who will converge on it (*Response Forces and Stations* §4.7, §8.3). The Route 91 Harvest festival in Las Vegas in October 2017, a gathering of more than 22,000, was planned with the police and a private medical provider and without the fire service, whose dispatch center did not know it was taking place, and the trauma supplies in its medical tent were exhausted within minutes (*Route 91 Harvest Shooting* F-5). The public-assembly class, when it is written, states the venue's obligations in full.

# Section 7 — Military-Action Extension Applicability

*[Skeleton.] Commercial Facilities is a recognized target of deliberate action for mass-casualty and psychological effect, and the two overlays sharpen the military-action case: an attack timed to a venue's disaster-support role strikes the population's refuge when it is most vulnerable, and an attack at a convergence node is a force multiplier that damages several sectors at once. The sector-level presumption toward MA applicability is strong for major mass-gathering venues, for facilities in their disaster-support role, and for convergence nodes. Application remains a per-facility determination under the mandatory two-factor test — geolocation and strategic value — with strategic value driven by occupancy, disaster-support role, and convergence exposure.*

# Section 8 — Facility Class Register and Inheritance Model

## 8.1 Inheritance Model

Facility-class instruments of every type in the DBA-CF series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework, and — as a consumer sector — inherit the physical design basis of their substrate from the energy, communications, transportation, and water series. Each instrument adds the facility-function-specific consequence chain and, where applicable, the disaster-support and convergence-node consequences.

## 8.2 Facility Class Register

The following table is the current facility-class register for the DBA-CF series. DBA-CF-PA-FC1 has issued as a working draft and the other function classes are at Anticipated status, with the disaster-support and convergence-node dimensions carried as overlays. Designators were ratified 2026-08-09 and sub-indexed by occupancy and convergence exposure.

| Designator | Function class (function-primary) | Governing characteristic | Status | Primary regulatory anchor |
|---|---|---|---|---|
| DBA-CF-PA-FC1 | Public assembly & mass-gathering venues — sports stadiums & arenas, performing-arts venues, theaters, convention centers, theme parks, outdoor events | mass-gathering life-safety; disaster-support (ESF-6) & convergence-node overlays | Issued (working draft) | Local codes; ISC; DHS soft-targets |
| DBA-CF-LR-FC2 | Lodging & real estate — hotels & motels, office buildings, multi-unit residential, mixed-use | occupancy life-safety | Anticipated | Local building codes; ADA; OSHA |
| DBA-CF-RT-FC3 | Retail & commerce — shopping centers & malls, retail, gaming/casinos, food service | commercial-gathering; economic-hub | Anticipated | Local codes; state gaming (where applicable) |
| DBA-CF-CorpCo† | Venue security & emergency coordination (coordination class) — private/event security and its public-safety interface | crowd management; response coordination | Anticipated (architecture) | DHS (SRMA); local public safety |
| (overlays) | Mass-care / disaster-support (venue-as-shelter, ESF-6 → DBA-ESS) · Convergence node (co-located CIKR → multi-sector consequence) | dual-role & geographic co-location | Anticipated (overlay) | FEMA ESF-6; cross-sector (TRAN/EN/COMM) |

*Designators were ratified 2026-08-09. Facility-class numbers are sequential across the series; CorpCo is a coordination class and carries no FC number, mirroring DBA-ES-CntlCo. Grounded in the CIKR_v16 sector map (Commercial Facilities Core Infrastructure and Operational Characteristics), extended with the disaster-support and convergence-node roles not present in the map.*

## Section 9 — Development Status and Open Items

This is the initial working draft of the DBA-CF sector framework and the sixteenth and final NIPP sector skeleton in the corpus. It establishes the standalone-series placement, the mass-gathering life-safety consequence, and the two roles the soft-target framing omits — the disaster-support (mass-care) role and the convergence-node role — the function-primary decomposition (public assembly, lodging and real estate, retail and commerce, and the CorpCo venue-security class) with an occupancy-and-convergence sub-index and disaster-support and convergence overlays, and the boundaries to Emergency Services and to the cross-sector interdependency analysis. The design-basis envelope (Section 4), the dual-DB detail (Section 5), and the MA applicability section (Section 7) are stubs to be developed. Open items at this version: develop the disaster-support and convergence-node consequence measures; and formalize the convergence-node hand-off to the interdependency analysis, now that all sixteen sectors are represented.

At v0.4 the planned-gathering interface with Emergency Services is stated (Section 6.3), closing the Commercial Facilities side of DBA-ESS-RE-11 §9 OI-3; its full statement is owed by the public-assembly class. At v0.5, the series' first facility class has issued: DBA-CF-PA-FC1 (*Public Assembly and Mass-Gathering Venues*), physical, and the three reference events it proposed are conferred at Section 9.1. It states the venue's and the promoter's side of the planned-gathering interface in full (*Public Assembly and Mass-Gathering Venues* §4.5), and the

statement this framework recorded as owed is made. It develops, for the public-assembly function, the three consequence treatments of Section 3.6 and the candidate bounding events of Section 4.1. What it notes to this framework is taken up at the framework's next conformance pass: the three chains, the six parameters and crowd density in particular, the occupancy tiers and the two overlays stated separately, the event categories it develops ahead of Section 4, the Safe Assembly and Sustained Refuge Conditions, its declared residuals and the consequence-basis conditionality check (*Public Assembly and Mass-Gathering Venues* §10.9). At v0.5 the Department of Homeland Security is named Sector Risk Management Agency, the current name of the role, at Sections 1.4 and 2.1, in the register and in the references (*Public Assembly and Mass-Gathering Venues* §10.5). Further open item: the three reference-event instruments of Section 9.1.

## 9.1 Conferred Reference Events

On the author's decision of 14 October 2026, this framework confers a reference-event designator on each of the three events its first facility class proposed. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), reference-event instruments are written and held by Project 11, DBA Reference Events; this framework keeps the conferral. None of the three instruments is yet written. The designators are numbered in the order the proposing class lists its events; the number records conferral, not issue. Each designator is sector-primary and single.

From *Public Assembly and Mass-Gathering Venues*, three events are conferred: DBA-CF-RE-1, the bombing at the Manchester Arena on 22 May 2017, read from the three volumes of the public inquiry; DBA-CF-RE-2, the crowd crush at the Astroworld festival in Houston on 5 November 2021; and DBA-CF-RE-3, the use of the Louisiana Superdome as the refuge of last resort in New Orleans during Hurricane Katrina in August and September 2005. The cyberattack on MGM Resorts in September 2023, which the plan for closing the seven sectors named for this sector, is held for *Lodging and Real Estate* (DBA-CF-LR-FC2) or *Retail and Commerce* (DBA-CF-RT-FC3) and is not conferred here. The destruction of the drama theater in Mariupol on 16 March 2022, which the class notes, is not conferred, and the shooting at the Route 91 Harvest festival remains the Emergency Services series' reference event (DBA-ESS-RE-11). The three instruments are owed, and they are Project 11's to write. *Public Assembly and Mass-Gathering Venues* states the three as proposed at its v0.1 and conforms to the conferral at its next revision.

This framework is still a skeleton at Sections 4, 5 and 7. The three instruments read their events back through the design basis of *Public Assembly and Mass-Gathering Venues*, which develops it ahead of the framework, and what they return to the framework is taken up at its next conformance pass.

# References

*[To be completed.] Governing and inherited instruments: FD-BL and determination annexes; DBA-MA Parent Framework (IAEA L1); DBA Series Gaps and Recommendations (§1.7, MA applicability); DBA-ESS (disaster-support / ESF-6 mass-care boundary); DBA-TRAN / DBA-EN / DBA-COMM / DBA-W (facility substrate and convergence-node co-located sectors). Regulatory: DHS (SRMA; securing soft targets and crowded places); local building and fire codes; ADA; OSHA; state gaming; FEMA (ESF-6 mass care). Grounding: CIKR_v16 sector map (Commercial Facilities Sector), extended with the disaster-support and convergence-node roles.* Planned-gathering interface: *Response Forces and Stations* (DBA-ESS-RF-FC2) §4.7 and §8.3; *The Route 91 Harvest Shooting of October 2017* (DBA-ESS-RE-11), cited by the short title *Route 91 Harvest Shooting* and finding number. Series instruments: DBA-CF-PA-FC1 (*Public Assembly and Mass-Gathering Venues*), the series' first facility class; DBA-CF-RE-1 to DBA-CF-RE-3, the conferred reference events of Section 9.1, their instruments owed to Project 11.

## Revision History

**0.5** October 2026 — T. Roxey. Section 9.1 is added and confers DBA-CF-RE-1 to DBA-CF-RE-3 on the three reference events that DBA-CF-PA-FC1 (Public Assembly and Mass-Gathering Venues) proposed, on the author's decision of 14 October 2026: the Manchester Arena bombing (2017), the Astroworld crowd crush (2021) and the Louisiana Superdome as refuge of last resort during Hurricane Katrina (2005). The MGM Resorts cyberattack (2023) is held for the lodging and retail classes. Each designator commits the series to a register row and a standalone instrument, none yet written; the instruments are Project 11's. The register at Section 8.2 enters the class as issued. Section 3.6 records the class's statement of the three consequence treatments. Section 9 records the class's issue, that the venue's side of the planned-gathering interface is now stated in full by the class, what the class notes to this framework, and the agency's name. The Department of Homeland Security is named Sector Risk Management Agency at Sections 1.4 and 2.1, in the register and in the references. References add the class and the three designators. No change to the taxonomy, the designators of the classes, the sub-index or the design basis. Revision-history entries are left as records.

**0.4** October 2026 — T. Roxey. Section 6.3 names a third boundary, the planned gathering, a shared interface with Emergency Services drawn by who acts on the author's decision of 7 October 2026: the venue's medical capacity and supplies sized for the surge, and the duty to bring every responding discipline into the event's plan, belong to this series; the response's plan, dispatch position and check-in belong to the Emergency Services series. Issued in the same pass as DBA-ESS-RF-FC2 v0.2 and from DBA-ESS-RE-11 Finding F-5. Section 9 and the references record it. Revision-history entries are left as records.

**0.3** October 2026 — T. Roxey. Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026. Section 5 states the scope of Bright Line placement by the determinism carve-out of FD-LD §4.3 rather than as any AI action, and states that determinism decides scope and not placement (P9-11, wording). Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its

sense: the decomposition tier reads function class, and the coordination layer reads coordination class (OI-15). No change to the design basis, the taxonomy or the register.

**0.2** August 2026 — T. Roxey. Designators ratified; coordination class resolved to CorpCo (private venue operations; disaster-support overlay hands off to Emergency Services); PA / LR / RT confirmed; disaster-support and convergence-node overlays confirmed.

**0.1** August 2026 — T. Roxey. Initial skeleton and the sixteenth and final NIPP sector framework, grounded in the CIKR_v16 sector map and extended beyond it. Established DBA-CF as a standalone sector series with the mass-gathering life-safety consequence and, going beyond the soft-target framing, two roles: the disaster-support / mass-care role (venue-as-shelter under ESF-6, a response asset whose value peaks during another sector's failure) and the convergence-node role (large urban venues stacking multiple co-located CIKR — the Madison Square Garden / Penn Station example — the first geography-organized consequence structure in the corpus and the bridge to the interdependency analysis). Adopted the function-primary decomposition (Public Assembly & Mass-Gathering / Lodging & Real Estate / Retail & Commerce / CorpCo venue-security) with an occupancy-and-convergence sub-index and disaster-support and convergence overlays; named the Emergency-Services (ESF-6) and cross-sector-convergence boundaries. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.